K-EL/01 Comprehensive Gap Closure Plan

Original-brief review, generation plan, acceptance gates and residual unknowns

1. Purpose and governing principle

This plan maps each requirement in KR-MECH-INT-001 to a concrete file/evidence item, states what has been generated, defines how to generate/check anything still missing, and prevents assumptions from being presented as verified facts. The challenge explicitly says this is conceptual work; it does not require physical prototype, purchases, detailed FEA or manufacturing-ready drawings.

2. Required deliverables versus generated evidence

Requirement	Brief asks	Evidence/generated output	Acceptance / outstanding
1. Calculation sheet	Static torque, target factor, output speed and motor speed/ratio.	Calculations.csv + Engineering Workbook.xlsx + report tables.	Formula and units checked against brief; input cells highlighted in workbook.
2. Component selection	One candidate motor + gearbox with official datasheets/product references.	Component_Research.csv + report with official URLs + Supplier_Clarifications.csv.	Shortlist only; request exact configured combination and loaded curve; not frozen until response.
3. CAD files	Elbow concept and simplified arm assembly; CAD may be FreeCAD/SolidWorks/Fusion/equivalent.	STEP + Build_K_EL01_STEP.py + FCMacro.	STEP re-import valid, 21 solids. Native .FCStd deferred: run macro inside FreeCAD; do not claim macro generated it here.
4. Drawing	One assembly drawing or clearly labelled CAD view.	STEP model plus concept drawing PDF/SVG/PNG.	Views and load paths included; concept dimensions only, no manufacturing tolerances.
5. Draft artwork	One isometric whole arm, enlarged elbow/exploded view, labels/load paths, key dimensions, torque/speed, trade-off note.	K_EL01_Concept_Drawing.pdf/.svg/.png.	One A3 landscape page; perform visual check at full resolution before packaging.
6. BOM	Main components, quantities, manufacturer/model if known.	BOM.csv + workbook BOM sheet.	Separates candidate/catalogue, original concept, conditional and TBD elements; no invented stock number for custom part.
7. Short report	~3–5 pages; assumptions, critical-thinking answers, two risks, rationale.	Kritva_K_EL01_Design_Report.docx/.pdf.	Target 3–5 PDF pages; DOCX rendered and visually inspected before final release.
Critical thinking Q1–Q6	Brief asks brief answers to each.	Report + speaker Q&A.	Answers checked against source framing.
Final review	10-minute presentation then 10-minute technical discussion.	PPTX/PDF + timed speaker notes + technical Q&A.	User must rehearse/deliver; live review itself cannot be performed by deliverable generation.
Original-source audit	Preserve original wording/requirements and show trace.	Original brief PDF included; Assignment_Traceability.csv.	Each original requirement maps to evidence/status; source remains unchanged.

3. Remaining technical closure plan (in order)

Gate	Actions	Evidence / outputs	Entry/exit rule
Gate 0 — Clarify system mission	Obtain duty cycle, trajectory, accelerations, max speed, dwell/reversals, impact/stop cases, ambient temperature, accuracy/backlash, life, cable force, load orientation and power-loss holding.	Project lead answers; revised requirement spec and signed assumption log.	Before motor/reducer design freeze; currently open because brief omits these inputs.
Gate 1 — Supplier configuration	Send exact motor 579165, GP52C ratio 91:1 candidate, encoder 512 CPT, controller 688770 concept to maxon. Ask compatible configured order number, actual torque-speed curve at 4095 rpm, output torque at 45 rpm, loaded efficiency, thermal limits, continuous/intermittent/duty curves, the 180 W versus nominal Tω power arithmetic, maximum input/output loads and regen needs. Do not treat the EC-i 0.321 rpm/mNm gradient as constant; Maxon support describes its iron-winding speed/torque characteristic as nonlinear.	Written vendor response, official datasheet revision, exact order code and performance point saved to Component_Research and Supplier_Clarifications.	Do not release part numbers or claim operational pass before response.
Gate 2 — Transmission interface	Select coupling hubs/bore/key, axial	Vendor-confirmed coupling assembly and	Current size 19/98ShA is a candidate

Gate	Actions	Evidence / outputs	Entry/exit rule
	retention, permitted misalignment, peak/reversing torque, service factor, torsional stiffness and speed. Confirm KTR hub compatible with 12 mm spindle and gearhead's actual keyed output.	interface sketch.	family only, not full set of order codes.
Gate 3 — CAD native file and assembly	Open STEP in FreeCAD (or accepted CAD), execute K_EL01_Split_Yoke_Concept.FCMacro, save .FCStd. Compare it to STEP; inspect object tree/units/orientation. Add proper constraints/joints only if supported by selected CAD package.	Native file saved from CAD; screenshot of assembly tree and isometric; export refreshed STEP.	FreeCAD unavailable in current authoring environment; user/FreeCAD workstation execution needed.
Gate 4 — Engineering detail	Specify shaft material/condition, shoulders, keyway, hub contact, retaining features, bearing fits, axial load/retention, seals, bolt grades/preload, flange bolt pattern, housing wall thickness, cable route, motor mount, tolerances and assembly access.	Released detailed drawings and calculation notes, reviewed by mechanical designer.	Not required for conceptual assignment; required before building hardware.
Gate 5 — Calculations	Include link inertia, motor rotor/gearhead inertia reflected through ratio, acceleration/deceleration, gravity across full range, impact/stop events, peak and RMS torque/current, output deflection, shaft combined stress with notch/ fatigue, key/hub contact, fasteners, carrier stiffness, bearing L10/oscillation and thermal equilibrium.	Updated calculation workbook; independent engineering review; explicit pass/fail thresholds derived from requirements.	Current static screens are inadequate for final release.
Gate 6 — Digital analysis	Perform CAD interference/motion checks; hand calculations first; FEA for yoke/shaft/flange only if useful, with mesh convergence, boundary-condition rationale and load cases. Use bearing/vendor tools for life and manufacturer reducer sizing tool.	Native model, assumptions, solver/settings, mesh convergence, plots and output tables.	FEA is explicitly not required in assignment; do not manufacture simulation output without running it.
Gate 7 — Safety and controls	Define safe state and power-loss hold. If gravity-load hold required, size normally engaged brake (including margin and release logic) or external support. Validate emergency stop, limits, encoder polarity, regenerative bus and controller derating.	Control concept, brake data or explicit excluded operating envelope, interface schematic and risk assessment.	No brake selected; supervised de-energized setup must remain externally supported.
Gate 8 — Bench verification (future, not assignment task)	Only after component configuration and detailed design: guarded, restrained test stand, no personnel under load, increment torque/speed; measure output torque/speed, current, temperature, backlash and stop response.	Calibrated test setup, protocol, raw logs, uncertainty statement, comparison against requirement thresholds.	No hardware purchased/manufactured; this gate is a future plan, not reported as performed.
Gate 9 — Release decision	Close every requirement and hazard; resolve supplier clarifications; sign drawings/BOM/configuration; archive revision, source, test evidence and review approval.	Design review record and controlled release package.	Current package stays "assignment concept only".

4. Suggested 5-day execution plan (aligned to brief duration)

Day	Activities	End-of-day evidence / acceptance
Day 1 — Requirements & sizing	Extract brief; establish source/assumption log; torque, power and speed math; identify missing duty/accuracy inputs.	Reviewed calculation sheet and clarification list; all units and assumptions traceable.
Day 2 — Drive shortlist	Compare motor/reducer candidates, official references, ratio, current/power/efficiency sensitivity and controller/encoder options.	Candidate stack and supplier questions; do not call it selected until configured point is validated.
Day 3 — Architecture & CAD	Create support-bearing load path, actuator envelope, simplified arm/wrist, assembly view, link lengths and assembly access review.	Valid STEP, editable builder source/macro, key dimensions and drawing export.
Day 4 — Risk, report and artwork	BOM, six Q answers, two major risks, one-page artwork, traceability, design/verification register.	3–5 page report, drawing PDF/SVG/PNG, risk and gap registers.
Day 5 — QA & presentation	Peer review calculations/files, render every PDF/DOCX page and slides, archive checksums, rehearse 10-minute talk and Q&A.	Integrity-checked release package and clear residual-risk statement.

5. Required acceptance checklists

- Calculation acceptance: inputs match brief; torque terms independently recomputed; units are visible on every equation; target factor labelled assumption; speed ratio and output power reconcile; sensitivity rows recalculate from formulas.
- Component acceptance: official part links reachable; revision/date captured; exact order codes and accessories verified; torque-speed-thermal point confirmed; no rated maximum or maximum efficiency represented as guaranteed operating point.
- CAD acceptance: file opens without warnings; global unit system stated; part coordinates and assembly placement documented; model geometry valid; no overlap/interference assumed merely from STEP validity; native editable source opened in target CAD and re-export round-trip compared.
- Drawing acceptance: whole-arm isometric and enlarged elbow/exploded view; labels distinguish candidate vendor envelopes from original parts; load-path arrows; COM radii and link lengths; torque and speed callout; at least one trade-off/open issue; text readable at 100%.
- Report acceptance: 3–5 pages; assumptions and limitations; concise BOM rationale; two risks with mitigations; all Q1–Q6 answered; source URLs; no broken placeholders; consistent spindle/coupling architecture.
- Presentation acceptance: 10-minute timed presentation, covering torque math, sizing, component evidence, load path, drawing/CAD, trade-offs and next validation; 10-minute discussion question prep; explicitly acknowledge unknowns.
- Package acceptance: generated source/editable file, PDF preview, named versions, index/hash, archive integrity; no intermediate QA image folders in final zip unless explicitly useful.

6. Current status and residual open items

At the time of this plan, the concept calculations, one-page artwork, candidate component references, STEP model, source CAD builder/macro, report, BOM and registers can be generated and checked. A native FreeCAD .FCStd file, supplier-confirmed drive combination, real thermal/motion proof, manufacturing detail, controlled safety design and physical-test results cannot responsibly be asserted from the provided assignment input. They have explicit routes and acceptance evidence above.